

What this result assumes

Single grid

One mesh level: 39680 cells across 3 regions. A discretisation error bar needs three grids. No numerical uncertainty is quoted for any row.

Steady points only

Each of the sixteen points is a separate converged steady state. Warm-up time and response to a load change are outside this result.

Radiation neglected

Radiation is off in all three regions. Heat leaves the body by conduction and forced convection alone. The real part has a radiative path; it is not represented.

Representative geometry and materials

The body is a representative motor-in-duct wedge. Conductivities are constant: 40, 167 and 0.026 W/mK for core, housing and air. It is not a particular product.

Turbulence is modelled, not resolved

The air side uses the k-omega SST closure. Convective heat transfer carries that model's error. No model-form uncertainty is quoted.

The peak is in the core

The hottest solid cell is in the core at 107.7 degC. That is 4.1 K above the housing peak. Margin to the 200 degC limit is computed on the core.

Mesh identity is checked, not assumed

All 16 cases byte-identical in fluid, housing and core – 48 of 48 region-case pairs.